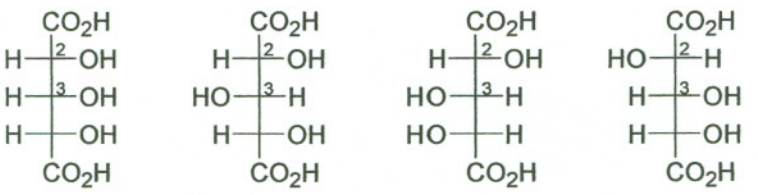
Organic cume
March 4, 2005

all questions equal weight

1. Define, using either a table or a diagram, the terms synperiplanar, antiperiplanar, synclinal, and anticlinal.
2. Define and give an example of atropisomers.
3. Define the C_n symmetry element and give an example (n can be any number you choose).
4. Give an example of molecules that are chiral by virtue of having each of the following: a stereogenic center, a stereogenic plane, and a stereogenic axis.
5. Explain the difference between configuration and conformation, and illustrate with an example.
6. Explain what is meant by homotopic and heterotopic faces of a trigonal atom?
7. What is a kinetic resolution?
8. Define the terms *l* and *u*.
9. Define the terms *lk* and *ul*.
10. Assign the stereochemical descriptors for the absolute configuration at C2 and C3 for each of the following molecules. (Hint: CIP rule 5: *R* precedes *S*).



What term is used to describe C3 of the meso compounds?